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$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{t=1}^T \hat{u}_t^2 = \min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t)^2$$

Solve (1) and (2) equations:

$$\frac{d \sum_{t=1}^T \hat{u}_t^2}{d \hat{\beta}_0} = 0, \text{ i.e., } \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t) = 0 \text{ --- (1)}$$

$$\frac{d \sum_{t=1}^T \hat{u}_t^2}{d \hat{\beta}_1} = 0, \text{ i.e., } \sum_{t=1}^T x_t (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t) = 0 \text{ --- (2)}$$

Then, we will have

$$\hat{\beta}_1 = \frac{\sum_{t=1}^T (x_t - \bar{x})(y_t - \bar{y})}{\sum_{t=1}^T (x_t - \bar{x})^2} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\begin{aligned} &= \frac{\beta_0 \sum_{t=1}^T (x_t - \bar{x})}{\sum_{t=1}^T (x_t - \bar{x})^2} + \frac{\beta_1 \sum_{t=1}^T (x_t - \bar{x}) x_t}{\sum_{t=1}^T (x_t - \bar{x})^2} + \frac{\sum_{t=1}^T (x_t - \bar{x})(u_t)}{\sum_{t=1}^T (x_t - \bar{x})^2} \\ &= 0 + \frac{\beta_1 \sum_{t=1}^T (x_t - \bar{x})(x_t - \bar{x})}{\sum_{t=1}^T (x_t - \bar{x})^2} + \frac{\sum_{t=1}^T (x_t - \bar{x})(u_t)}{\sum_{t=1}^T (x_t - \bar{x})^2} = \beta_1 + \frac{\sum_{t=1}^T (x_t - \bar{x})(u_t)}{\sum_{t=1}^T (x_t - \bar{x})^2} \end{aligned}$$

Conditional on X

$$Var(\hat{\beta}_1) = Var(\beta_1) + Var\left(\frac{\sum_{t=1}^T (x_t - \bar{x})(u_t)}{\sum_{t=1}^T (x_t - \bar{x})^2}\right)$$

We can assume $\bar{x} = 0$ for the sake of simplicity. Note that our final answer that the wrong formula underestimates the true variance estimate will still hold even if $\bar{x} \neq 0$.

$$Var(\hat{\beta}_1) = 0 + Var\left(\frac{\sum_{t=1}^T (x_t u_t)}{\sum_{t=1}^T x_t^2}\right)$$

$$Var(\hat{\beta}_1) = \frac{1}{(\sum_{t=1}^T x_t^2)^2} Var\left(\sum_{t=1}^T (x_t u_t)\right)$$

$$Var(\hat{\beta}_1) = \frac{1}{(\sum_{t=1}^T x_t^2)^2} Var(x_1 u_1 + x_2 u_2 + x_3 u_3 + \cdots + x_T u_T)$$

Remark (1)

What is $Var(x_1u_1 + x_2u_2 + x_3u_3 + \cdots + x_Tu_T)$?

MH1820 and HE2003 Recap: Using 2 and 3 random u variables case to illustrate the variance formula.

$$(a) Var(x_1u_1 + x_2u_2) = x_1^2Var(u_1) + x_2^2Var(u_2) + 2x_1x_2Cov(u_1, u_2)$$

$$(b) Var(x_1u_1 + x_2u_2 + x_3u_3) = x_1^2Var(u_1) + x_2^2Var(u_2) + x_3^2Var(u_3) \\ + 2x_1x_2Cov(u_1, u_2) + 2x_1x_3Cov(u_1, u_3) + 2x_2x_3Cov(u_2, u_3)$$

There are two possible cases: Case A and Case B.

Case A: TS 5 holds, i.e., u terms are uncorrelated. The answer below shows that the covariance terms are 0 due to no serial correlation.

$$Var(\hat{\beta}_1) = \frac{1}{(\sum_{t=1}^T x_t^2)^2} \left\{ \sum_{t=1}^T x_t^2 Var(u_t) + 2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} Cov(u_t, u_{t+j}) \right\}$$

Case B: TS 5 is violated, i.e., u terms are serially uncorrected. Therefore, covariance terms are nonzero in the $Var(\hat{\beta}_1)$.

$$Var(\hat{\beta}_1) = \frac{1}{(\sum_{t=1}^T x_t^2)^2} \left\{ \sum_{t=1}^T x_t^2 Var(u_t) + 2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} Cov(u_t, u_{t+j}) \right\}$$

END of Remark (1)

Since Question 1 talks about the situation where TS 5 is violated. We have Case B.

$$Var(\hat{\beta}_1) = \frac{1}{(\sum_{t=1}^T x_t^2)^2} \left\{ \sum_{t=1}^T x_t^2 Var(u_t) + 2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} Cov(u_t, u_{t+j}) \right\}$$

Remark (2)

For $2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} \text{Cov}(u_t, u_{t+j})$, if you are not clear about this math expression, then you can ignore this expression and try to write all the covariance terms one by one as you like, i.e.,

$$2x_1x_2\text{Cov}(u_1, u_2) + 2x_1x_3\text{Cov}(u_1, u_3) + 2x_2x_3\text{Cov}(u_2, u_3) + \cdots + 2x_{T-1}x_T\text{Cov}(u_{T-1}, u_T)$$

But, here I use $2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} \text{Cov}(u_t, u_{t+j})$ as the expression is neat and elegant.

END of Remark (2)

$$\text{Var}(\hat{\beta}_1) = \frac{1}{(\sum_{t=1}^T x_t^2)^2} \left\{ \sum_{t=1}^T x_t^2 \text{Var}(u_t) + 2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} \text{Cov}(u_t, u_{t+j}) \right\}$$

$$\text{Var}(\hat{\beta}_1) = \frac{1}{(\sum_{t=1}^T x_t^2)^2} \left\{ \sum_{t=1}^T x_t^2 \sigma^2 + 2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} \rho^j \sigma^2 \right\}$$

Remark (3)

$$Var(\hat{\beta}_1) = \frac{1}{(\sum_{t=1}^T x_t^2)^2} \left\{ \sum_{t=1}^T x_t^2 Var(u_t) + 2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} Cov(u_t, u_{t+j}) \right\}$$

$$Var(\hat{\beta}_1) = \frac{1}{(\sum_{t=1}^T x_t^2)^2} \left\{ \sum_{t=1}^T x_t^2 \sigma^2 + 2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} \rho^j \sigma^2 \right\}$$

Why they are same? You can check a few special cases and find the pattern.

$$Cov(u_t, u_{t+1}) = Cov(u_t, \rho u_t + e_t) = \rho Cov(u_t, u_t) = \rho \sigma^2$$

$$\begin{aligned} Cov(u_t, u_{t+2}) &= Cov(u_t, \rho u_{t+1} + e_{t+1}) = Cov(u_t, \rho^2 u_t + \rho e_{t+1} + e_t) \\ &= \rho^2 Cov(u_t, u_t) = \rho^2 \sigma^2 \end{aligned}$$

It's easy to conclude that it is also true that $Cov(u_t, u_{t+j}) = \rho^j \sigma^2$

End of Remark (3)

$$Var(\hat{\beta}_1) = \frac{1}{(\sum_{t=1}^T x_t^2)^2} \left\{ \sum_{t=1}^T x_t^2 \sigma^2 + 2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} \rho^j \sigma^2 \right\}$$

$$Var(\hat{\beta}_1) = \frac{\sigma^2 \sum_{t=1}^T x_t^2}{(\sum_{t=1}^T x_t^2)^2} + 2 \times \frac{\sigma^2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} \rho^j}{(\sum_{t=1}^T x_t^2)^2}$$

Clearly, most of the time

$$\begin{aligned} Var(\hat{\beta}_1) &= \frac{\sigma^2}{\sum_{t=1}^T x_t^2} + 2 \times \frac{\sigma^2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} \rho^j}{(\sum_{t=1}^T x_t^2)^2} \\ &> \frac{\sigma^2}{\sum_{t=1}^T x_t^2} \text{ (The conventional formula where } \bar{x} = 0) \end{aligned}$$

So, below is the correct variance formula when TS 5 is violated

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{t=1}^T x_t^2} + 2 \times \frac{\sigma^2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} \rho^j}{(\sum_{t=1}^T x_t^2)^2}$$

Remark (4)

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{t=1}^T x_t^2} + 2 \times \frac{\sigma^2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} \rho^j}{(\sum_{t=1}^T x_t^2)^2}$$

This is the correct variance formula. If No Serial Correlation assumption is violated, we use this correct formula. The **green part** is often a POSITIVE value as x time series variables are often positively correlated.

If No Serial Correlation assumption holds, then we **don't need this term as $\rho = 0$**

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{t=1}^T x_t^2} + 2 \times \frac{\cancel{\sigma^2 \sum_{t=1}^{T-1} \sum_{j=1}^{T-t} x_t x_{t+j} \rho^j}}{\cancel{(\sum_{t=1}^T x_t^2)^2}} = \frac{\sigma^2}{\sum_{t=1}^T x_t^2}$$

End of Remark (4)

2.

Since the question says

$$(1) y_t = a_0 + a_1 y_{t-1} + u_t, \text{ and } u_t = \rho u_{t-1} + e_t$$

Then, clearly, it is also true that

$$(2) y_{t-1} = a_0 + a_1 y_{t-2} + u_{t-1}, \text{ and } u_{t-1} = \rho u_{t-2} + e_{t-1}$$

If we “replace” u_t by $\rho u_{t-1} + e_t$, then we substitute away u_{t-1} based on the equation (2).

$$y_t = a_0 + a_1 y_{t-1} + u_t$$

$$\Rightarrow y_t = a_0 + a_1 y_{t-1} + (\rho u_{t-1} + e_t)$$

$$\Rightarrow y_t = a_0 + a_1 y_{t-1} + [\rho(y_{t-1} - a_0 - a_1 y_{t-2}) + e_t]$$

$$\Rightarrow y_t = (1 - \rho)a_0 + (a_1 - \rho a_1)y_{t-1} - \rho a_1 y_{t-2} + e_t$$

The u_t is no longer found in the equation and only (good) e_t in the final equation. The serial correlation is no longer the problem.